Ahmed Ayman Al-Hofy

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https://github.com/AhmedAyman4 | https://www.linkedin.com/in/ahmed-alhofy/ | View my Portfolio!

Summary

Information Systems student, AI Engineer, and Data Scientist focused on RAG, NLP, and data automation. Experienced in building end-to-end agents and predictive models that solve real-world problems. Passionate about bridging the gap between raw data and functional AI products.

Education

Bachelor's Degree, Faculty of Computers & Artificial Intelligence – University of Sadat City — Graduating June 2026

CGPA: 3.3, Major: Information Systems.

Experience

AI Engineer | Konecta · Internship — New Cairo, Egypt | Jul 2025 - Present

- Collected, cleaned, and structured data using Python and web scraping tools to support reliable analysis and modeling.
- Built and optimized machine learning models (classification, clustering, image recognition) and implemented RAG-based chatbots with embeddings, document indexing, and multi-turn conversations.
- Automated data and AI workflows using n8n, improving efficiency and reducing manual processing.

Data scientist | Digital Egypt Pioneers Initiative - DEPI · Internship — Alexandria, Egypt | Oct 2024 – May 2025

- Learned to analyze data using Python, SQL, and analytics tools to uncover trends, opportunities, and actionable business insights.
- Learned to build, deploy, and track ML models using Python, Scikit-learn, and MLflow, enhanced with AI and prompt engineering for seamless integration.

Intern | CIB Egypt · Internship — Cairo, Egypt | Jul 2024 - Jul 2024

- Developed data, financial, and entrepreneurial skills to strengthen problem-solving and decision-making.

Skills

- **Technical:** Python (ML/DS), SQL, Bash, HTML/CSS, LangChain, LangGraph, LLMs (Hugging Face), Prompt Engineering, RAG, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Web Scraping (Playwright, BeautifulSoup).
- **Tools & Skills:** AWS, Google Cloud Platform (GCP), Vector Databases, Stats, Tableau, Power BI, Excel, Docker, Git, OOP, Problem-Solving, Communication.

Projects

Retail Analytics Copilot (Hybrid RAG + SQL Agent)

- Built a Dockerized Retail Analytics Copilot in Streamlit featuring a stateful LangGraph agent that routes user queries to SQL, RAG, or HYBRID paths using chain-of-thought reasoning and keyword heuristics; implemented a self-correcting, schema-constrained SQL pipeline with DSPy, regex sanitization, and error-driven retries; and fused BM25 document retrieval with natural language constraint extraction and a synthesizer to generate cited, contextual answers.

Semantic Book Recommender:

- **Accomplished a 40% improvement in recommendation relevance** (compared to keyword-based search) by engineering a semantic search engine using **Sentence-Transformers** for **vector embeddings** and **Zero-Shot classification** to map books to specific emotional and categorical tags.
- **Developed and deployed a production-ready full-stack application** with a **Next.js** frontend and **FastAPI** backend, achieving seamless multi-environment scalability through **Docker** containerization and automated **GitHub Actions** workflows.
- **Optimized data discovery and user insight** by preprocessing a dataset of 7,000+ books through comprehensive **EDA**, **NLP-driven sentiment analysis**, and the creation of custom-tagged datasets in **Jupyter Notebooks**, enabling users to find titles based on mood and complex semantic queries.

Customer Segmentation:

- Performed customer segmentation on wholesale spending data using K-Means (selected via Elbow and Silhouette analysis), Hierarchical, DBSCAN, and MeanShift clustering supported by log-transformation, PCA for dimensionality reduction, and rigorous evaluation metrics; derived six actionable customer profiles (e.g., Fresh-First, Bulk Buyers, Household Essentials) to inform targeted marketing, inventory planning, and high-value client prioritization.

Data Scraper (Real Estate & E-Commerce):

- Built robust Python scrapers using Playwright and BeautifulSoup to extract 1,400+ real estate listings from Nawy.com and 250+ gaming laptop products from Noon.com; automated infinite scrolling and pagination with error-resilient parsing, structured data into pandas DataFrames, and saved versioned CSVs, reducing manual data collection effort by over 90%.

Automated Document Processing:

- Built an AI-powered document processing system using OCR (PyTesseract, pdfplumber) and LangChain with Google's Gemini LLM to classify and extract structured data from invoices, purchase orders, and approvals, reducing manual processing by 80%; deployed a Streamlit/Flask app with real-time extraction and MongoDB-backed semantic search for interactive querying and duplicate detection.

Courses And Certifications

- SAP Certified Associate - SAP Generative AI Developer Issued by SAP — Sep 2025
- AWS Certified AI Practitioner Issued by Amazon Web Services Training and Certification — Apr 2025
- Oracle Cloud Infrastructure 2025 Certified Al Foundations Associate Issued by Oracle — Jun 2025
- Data Scientist Associate Issued by Data Camp — Aug 2024
- IBM SkillsBuild Data Analytics Certificate Issued by IBM SkillsBuild — Jun 2025
- AWS Academy Graduate - AWS Academy Machine Learning for Natural Language Processing — Aug 2024
- AWS Academy Graduate - AWS Academy Cloud Foundations — Jun 2024
- AWS Academy Graduate - AWS Academy Data Engineering — Jun 2024
- Applied Data Science with Python - Level 2 Issued By IBM — May 2024
- Google Data Analytics Professional Certificate Issued by Coursera — Jul 2024
- The Machine Learning Process A-Z – Issued By 365 Data Science — Nov 2024
- McKinsey Forward Program - Issued by McKinsey & Company — Dec 2024